8.012 Problem Set 10

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Problem 6

Kleppner & Kolenkow, Problem 9.3

9.3 A particle moves in a circle under the influence of an inverse cube law force. Show that the particle can also move with uniform radial velocity, either in or out. (This is an example of unstable motion. Any slight perturbation to the circular orbit will start the particle moving radially, and it will continue to do so.) Find θ as a function of r for motion with uniform radial velocity.

Let Mass of the particle be m.

Inverse Cube Law is a Central Force here. So Angular Momentum L is conserved here. Let the Force be

$$-\frac{k}{r^3} \qquad \text{Here } k \text{ is a constant}$$

Angular Momentum Conservation gives

$$mr^2\dot{\theta} = \text{constant}$$

$$\implies \dot{\theta} = \frac{c}{r^2} \qquad \text{Here } c \text{ is a constant}$$

Force Equation gives

$$m\ddot{r} - mr\dot{\theta}^2 = -\frac{k}{r^3}$$

$$\implies \ddot{r} = \left(c^2 - \frac{k}{m}\right)\frac{1}{r^2}$$

For a Motion with Uniform Radial Velocity to exist i.e. ($\ddot{r} = 0$), $\ddot{r} = 0$ should be a permissible in the Equation derived above.
Putting $\ddot{r} = 0$,

$$\left(c^2 - \frac{k}{m}\right)\frac{1}{r^2} = 0 \implies c^2 = \frac{k}{m}$$

This is acceptable since c, k and m are all constants and such a case can exist.

Hence proved,

There can be a case when the Particle can move with Uniform Radial Velocity.
And since $\ddot{r} = 0$ in this case, the motion is Unstable cause any Radial Velocity introduced by Disturbance will remain present all the time

$$\ddot{r} = 0 \implies r = vt + r_0 \qquad [r_{t=0} = r_0; \dot{r} = v]$$

Using $c^2 = \frac{k}{m}$ in the Angular Momentum

$$\dot{\theta} = \sqrt{\frac{k}{m}} \frac{1}{r^2} = \sqrt{\frac{k}{m}} \frac{1}{(vt + r_0)^2}$$

$$\implies \theta = \sqrt{\frac{k}{m}} \int_0^t \frac{dt}{(vt + r_0)^2} \implies \theta = -\sqrt{\frac{k}{m}} \frac{1}{v(vt + r_0)}$$

Therefore,

$$\boxed{\theta(r) = -\frac{\sqrt{k}}{vr\sqrt{m}}}$$

v is the Radial Velocity.

Problem 8

Kleppner & Kolenkow, Problem 9.6

9.6 A particle of mass m moves under an attractive central force Kr^4 with angular momentum l. For what energy will the motion be circular, and what is the radius of the circle? Find the frequency of radial oscillations if the particle is given a small radial impulse.

$$l = mvr \implies v = \frac{l}{mr}$$

For Circular Motion,

$$-\frac{mv^2}{r} = -Kr^4 \implies \frac{l^2}{mr^3} = Kr^4$$

$$\implies r = \left(\frac{l^2}{mK}\right)^{\frac{1}{7}} \qquad \frac{l^2}{m} = Kr^7$$

Potential Energy,

$$-\frac{dU}{dr} = -Kr^4 \implies U = \frac{K}{5}r^5 = \frac{K}{5}\left(\frac{l^2}{mK}\right)^{\frac{5}{7}}$$

Kinetic Energy,

$$T = \frac{mv^2}{2} = \frac{1}{2}\frac{l^2}{mr^2} = \frac{Kr^5}{2} = \frac{K}{2}\left(\frac{l^2}{mK}\right)^{\frac{5}{7}}$$

Total Energy is given by

$$E = T + U = \left(\frac{K}{2} + \frac{K}{5}\right)\left(\frac{l^2}{mK}\right)^{\frac{5}{7}}$$

Therefore, Energy required for Circular Motion is given by

$$\boxed{\frac{7K}{10}\left(\frac{l^2}{mK}\right)^{\frac{5}{7}}}$$

$$U_{\text{Effective}} = \frac{l^2}{2mr^2} + \frac{Kr^5}{5}$$

$$\implies \frac{d^2U_{\text{Effective}}}{dr^2} = \frac{3l^2}{mr^4} + 4Kr^3 = 7Kr^3$$

Now,

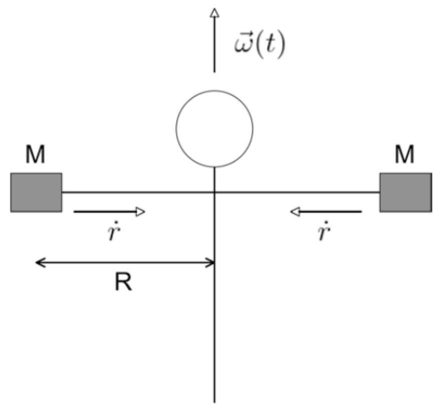
$$\omega = \sqrt{\frac{1}{m}\frac{d^2U_{\text{Effective}}}{dr^2}} = \sqrt{\frac{7Kr^3}{m}}$$

Therefore, the Angular Frequency for Radial Oscillations is given by

$$\boxed{\omega = \sqrt{\frac{7K}{m}}\left(\frac{l^2}{mK}\right)^{\frac{3}{14}}}$$

Problem 10

10. **Spinning up Figure Skater.**



(a) The derivation of the rotational fictional forces we did in class (coriolis and centrifugal) assumed that the spin vector $\vec{\omega}$ was constant in time. Relax this constraint and derive an additional fictional force that depends on the $\dot{\vec{\omega}}$ (this is known as the azimuthal force).

(b) We found earlier that the spin up of an ice skater as she brings her hands in can be attributed to conservation of angular momentum (for different moments of inertia). Let's examine this in the skater's frame of reference. Assume our skater has hands of mass M extended initially at a radius R from her body, attached to effectively massless arms and an effectively massless body (think: spherical cow). She is initially rotating with spin vector $\vec{\omega}$ pointing upwards. Our skater moves her hands slowly toward the center of her body at a rate $\dot{r}$. Based on the fact that the skater does not spin up in her own frame of reference (always true!), and considering all fictional forces in her rotating frame of reference, derive an expression for $\dot{\omega}$ in terms of ω, R and $\dot{r}$.

(c) Derive this same expression by examining the evolution of angular momentum in the inertial reference frame.

(d) If the skater pulls her hands in to half their original radius, by how much does she spin up?

(a)

In a rotating reference frame with time-dependent angular velocity $\boldsymbol{\omega}(t)$, the fictitious accelerations acting on a particle include the Euler (azimuthal) term

$$\vec{a}_{\text{Euler}} = -\dot{\boldsymbol{\omega}} \times \vec{r}$$

Multiplying by the mass M, the corresponding fictitious force is

$$\boxed{\vec{F}_{\mathbf{az}} = -M\,\dot{\boldsymbol{\omega}} \times \vec{r}}$$

This force acts tangentially and is responsible for changes in the angular speed of the system in the rotating frame.

(b)

We analyze the motion in the skater's rotating frame. Since the skater does not observe herself spinning up in her own frame, the net torque about the rotation axis must vanish.

For one mass at distance r, the magnitude of the azimuthal force is

$$F_{\mathrm{az}} = Mr\dot{\omega}$$

The torque about the axis due to this force is

$$\tau = rF_{\mathrm{az}} = Mr^2\dot{\omega}$$

With two identical masses, the total torque is

$$\tau_{\mathrm{total}} = 2Mr^2\dot{\omega}$$

Setting the net torque equal to zero,

$$\frac{d}{dt}(2Mr^2\omega) = 0$$

Thus,

$$2Mr^2\omega = \text{constant}$$

Using the initial condition,

$$2MR^2\omega_0 = 2Mr^2\omega$$

we obtain

$$\boxed{\omega = \omega_0 \frac{R^2}{r^2}}$$

(c)

In the inertial frame, no external torque acts on the skater–mass system. Therefore, angular momentum about the vertical axis is conserved:

$$L = 2Mr^2\omega = \text{constant}$$

Using the initial configuration,

$$2MR^2\omega_0 = 2Mr^2\omega$$

which again gives

$$\boxed{\omega = \omega_0 \frac{R^2}{r^2}}$$

(d)

If the skater pulls her hands in to half their original radius,

$$r = \frac{R}{2}$$

then

$$\omega = \omega_0 \frac{R^2}{(R/2)^2} = 4\omega_0$$

$$\boxed{\begin{aligned} \vec{F}_{\mathbf{az}} &= -M\,\dot{\boldsymbol{\omega}} \times \vec{r} \\ \omega &= \omega_0 \frac{R^2}{r^2} \\ \omega_f &= 4\omega_0 \quad \textbf{for } r = \frac{R}{2} \end{aligned}}$$